

Distress Screaming vs Joyful Screaming: An Experimental Analysis on Both the High Pitch Acoustic Signals to Trace Differences and Similarities

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Abstract— A shout, scream or yell is a loud vocalization in which air is passed through the vocal folds with greater force than is used in regular or close-distance vocalization. This process can be performed by any creature possessing lungs, including humans. A scream is a natural action, with a strong emotional aspect, like fear, pain, surprise, joy, anger, and others. But there is a difference between the scream in fear or pain and joy or surprise. This research is based on the study of such parameters those separates a scream of distress from a hoot in the joy of a person specifically women, having objective to design and develop security devices based on women emotions to provide them security. The present study has used high-reliability audio recordings to capture the acoustic features of both high pitch signals of women. Results indicated that perceptually categorized both the signals have distinct acoustic features. Distress scream reflect a slightly higher level of intensity than joyful scream. Moreover, the duration of the signal plays a vital role.

Keywords— *distress acoustic signal, hooting, screaming, scream in joy, women scream, differences, similarities*

I. INTRODUCTION

Screams are a means of expressing various states of emotion including anger, distress, fear, or joy, etc. Distress or panic scream is considered as a non-linguistic but sustained and loud vocalization. Similarly, a joyful scream is referred to as hooting and having similar properties of sustainability and loudness. For our future studies, we differentiate a panic scream from the scream made in joy. Human sounds generated through the oral and nasal cavity can generally be categorized into two main categories: (i) vocalizations or speech and (ii) non-vocalizations or non-speech. Non-speech sounds include screams, whistles, hoots, coughs, snore, laugh, sneeze, hiccups and sounds generated by using the different forms of mouth or tongue, etc., none of which contain any systematic organization of alphabets. This paper focuses on the analysis and categorization of screams under stress, fear, pain, and joy with the help of a few parameters, which reflect a portion of the class of non-speech sounds. These acoustic parameters help to separate a distress scream from a joyful one of a person specifically women. We considered these two because there is always a requirement to distinguish high pitch signals from one another. Work-related to screaming signal processing has been considered in [1, 2, 3, and 4]. David Poeppel is a researcher, an NYU professor of psychology and neural science and director of the Max Planck Institute for Empirical Aesthetics in Germany, who has done valuable

research on human scream and gives them a characteristic acknowledged as "roughness" [1].

In [2], a real-time scream detection technique was described. It was based on a combination of stability of log energy detection, high pitch analysis and compact MFCCs across frames using an SVM technique. In [3], by using multivariate adaptive regression splines (MARS) and support vector machines (SVM) human sounds including cough, scream, laugh, and snore was distinguished and classified. In [4], the performance of acoustic features including audio spectral flatness (ASF), MFCCs, linear-prediction cepstral coefficient (LPCC) and Mel-spectrum was compared for scream detection using low power SVM classifiers. James Green and Pamela G. Whitney are Researchers from the Department of Psychology, University of Connecticut and Michael Potegal is from Departments of Pediatrics & Neurology, University of Minnesota Medical Center. They had done a similar kind of study for children to find their emotion tantrums by taking samples of cries/yell and screams [5]. The authors have applied the proposed method for the spontaneous emotion recognition from speech using formant analysis in [6]. In [7], the authors analyzed the distinctiveness of scream of male speakers with the help of fundamental frequency analysis and Frame energy distribution analysis. They had also reported the impact of screaming on the performance of text independent speaker recognition systems which are observed that speaker recognition systems are not reliable when tested with scream. In [8], same authors done the detailed study to find the impact of screams on speaker recognition systems and concluded that traditional speaker recognition based on the Gaussian mixture models-universal background model framework is unreliable when evaluated with screams. In [9], the authors presents an audio-based video surveillance system which automatically detects abnormal audio events in a public square, for example screams or gunshots, and also pinpoint the position of the acoustic source, in such a way that a video-camera is steered accordingly. The system uses GMM classifiers to perform the job. The Experimental results show that the system can detect events with an accuracy of 93% at a false rejection rate of 5% when the SNR is 10dB, at the same time the source direction can be estimated with a precision of one degree. In [10] authors have proposed a development of a mobile phone's smart back panel to help specifically women in case of emergency. The system works with the audio vent such as a loud scream of the women. All of these studies related to scream analysis

have explored the domain of auditory analysis for audio observation. However, screams have not yet been studied to deduce useful information that could differentiate it with other high pitched and loud sounds without using classifiers. This is the first detailed study of the analysis of scream sound along with the different parameters which make a differentiation between scream in panic and joy. Although these are the initial results but observed as important for the future study.

The rest of the paper is organized as follows: Section. II describe the methodology followed for the experimental analysis. In the subsequent section, a detailed analysis of both the vocalization in terms of some important parameters is presented. Section IV presents concluding comments and directions for future research.

II. METHODOLOGY

To analyze differences in acoustic structure for these two classes, an acoustic analysis on a set of audio recordings was performed for ten voluntarily participated speakers (all were females from 18-26 age group). Further, each recording has been analyzed to find out the values for the following parameters (1) F0 contour and maximum frequency content; (2) Sound intensity in terms of loudness; (3) Formant location and (4) Duration of the signal.

A. Participants

For the initial study, volunteer participation has been chosen as a primary tool to record distress signal and a joyful scream, and for this purpose, female students from different colleges and institutions have been approached. Few essential health parameters have been considered while selecting students to make sure that the signal they generate will be accurate for the analysis. These parameters are:

1. Whether the student has any chronic throat infection/issue?
2. Whether the student has any nasal problem ex: sinus?
3. Any other medical issue which can stop you to scream loud?

A total of ten participants are considered for the experiment. All are from the age group of (18-26).

B. Recordings

Audio recordings have made using X-NUCLEO-CCA02M1 board which consists of two MP34DT01-M digital MEMS microphones onboard. The microphone is found to have a flat frequency response up to 10 kHz and acoustic overload point is 120dB SPL. Moreover, it supports Omni-directional sensitivity which means the signals can be captured from any direction. The sampling rate was 44 kHz, with 16-bit PCM. Open theater in university campus has been used for data acquisition. The students have been directed to make loud vocalization for both types of acoustic signals. They are told to feel the pain of danger and then make an attempt. Similarly, for joyful screams, students need to feel the joy first and then make an attempt. In all recordings, the distance from the mouth of a subject to a microphone is 1.8 meter.

Note: A scream has an analogical utterance of “aaaaaaaaaaaaaaaaaaaaa”, whereas the joyful scream is

similar to the tone “woooooo” or hooooooooo” or “yeeeeeee” etc. Fig.1 depicts two images of recording the samples from each category.

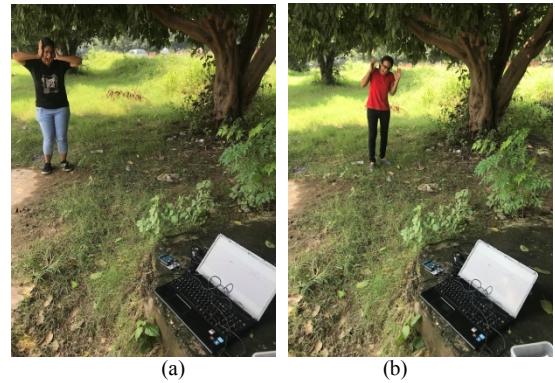


Fig. 1. (a) Panic scream, (b) Joyful scream.

C. Coding

The digital files representing each tantrum are coded using **Praat**; a program designed for acoustic analysis of speech [11]. The basic unit of coding is the girl's vocalization, defined as any perceptible activity of the vocal tract emitted during the course of a single respiratory expiration.

D. Analysis

This is the basic analysis and has been carried out to describe the optimal linear combination of acoustic features for separating the vocalization categories. The following measurements have been taken to extort useful information. 1) F0 contour and Maximum frequency content 2) Formant analysis 3) Sound intensity 4) Duration of the signal. The Spectrogram analysis has been done using a short-time Fourier transform method.

III. RESULTS AND DISCUSSION

This section provides a detailed analysis of changes information and attributes of scream vocalization during danger/pain relative to scream vocalization during joy.

A. Fundamental Frequency or F0 Contour

It represents emotion in voice data/speech. The fundamental frequency is considered to be one of the important parameters found to vary under different vocalization conditions. F0 contours for joyful scream and panic or scream under dangerous scenarios are computed from signals using autocorrelation algorithm. The F0 range is set to 50–600 Hz. It has been observed that there is no significant difference with mean computed across all speakers which further indicated that there is not much difference in the fundamental frequency of both the signals in this particular window. While the standard deviation for distress scream increased significantly as compared to joyful scream. It has also found that the maximum frequency of distress scream is up to 5 kHz but it's half in case of a joyful scream that is up to 2.5 kHz. Table I shows F0 comparison for both the vocalization across participants.

TABLE I. F0 COMPARISON FOR BOTH THE VOCALIZATION ACROSS PARTICIPANTS

Participant	Joyful Scream (Hz)		Distress Scream(Hz)	
	Mean F0	Standard deviation	Mean F0	Standard deviation
P1	444.44	90.54	364.71	149.134
P2	421.67	86.67	404.97	127.961
P3	393.17	82.56	358.241	139.983
P4	470.90	67.90	427.978	120.745
P5	401.11	69.45	448.622	87.438
P6	467.67	83.56	427.409	92.781
P7	330.77	79.67	483.789	115.337
P8	392.89	56.78	330.382	81.817
P9	400.78	89.33	359.221	129.980
P10	428.56	75.45	407.178	128.245

B. Sound Intensity

It referred to the amount of energy flowing a specific per unit time through a specific unit area that is perpendicular to that from where the sound waves are traveling. It can be measured in units of energy or work—e.g., micro joules per second per square centimeter—or in units of power, as microwatts per square centimeter. Correspondingly the sound level is measured in decibels. Table II shows the comparison of intensity in terms of sound level for both the vocalizations across participants.

TABLE II. COMPARISON OF INTENSITY IN TERMS OF SOUND LEVEL FOR BOTH THE VOCALIZATIONS ACROSS PARTICIPANTS

Participant	Joyful scream (dB)		Distress scream (dB)	
	Maximum intensity	Minimum intensity	Maximum intensity	Minimum intensity
P1	77.67	48.99	92.95	55.64
P2	75.23	36.90	92.99	54.17
P3	69.56	44.78	92.92	55.43
P4	72.67	39.89	91.22	47.08
P5	81.34	50.23	87.70	62.78
P6	80.56	49.77	86.91	66.58
P7	71.89	50.01	89.09	74.33
P8	66.45	42.87	86.67	56.54
P9	61.55	46.56	89.78	60.89
P10	53.90	30.82	90.56	61.66

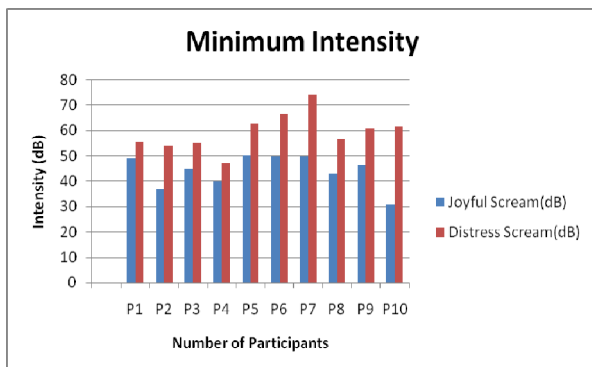


Fig. 2. Minimum intensity in both vocalizations.

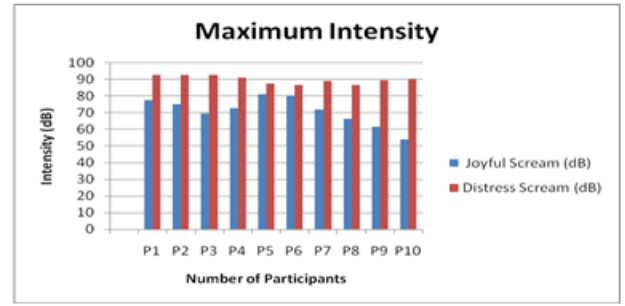


Fig. 3. Maximum Intensity in both vocalizations.

It is clear from the Fig. 2 and Fig. 3 that there is a significant amount of differentiation in terms of intensity of both the vocal signals. Distress scream has an intensity range of 47dB to 93dB whereas joyful scream depicts low sound level as compared to distress signal.

C. Formant Locations

A formant is a concentration of acoustic energy around a particular frequency in the speech wave come from the vocal tract. The air inside the vocal tract vibrates at different pitches depending on its size and shape of the opening. Formant measurements are one of the most significant factors to objectively investigate differences among vowels generated by different speakers. Generally, the distress scream falls in the //aaaaa// vowel vocalization whereas joyful scream has symbolically //uhoooo// or //uuuuuu// for whoo pronunciation. There are several formants, each at a different frequency, roughly one in each 1000Hz band. Each formant corresponds to a resonance in the vocal tract. These can be seen very clearly in a wideband spectrogram, where they are displayed as dark bands. The darker a formant is reproduced in the spectrogram, the stronger it is. Formants were extracted using LPC coefficients with the algorithm by Burg, as given by [12][13].

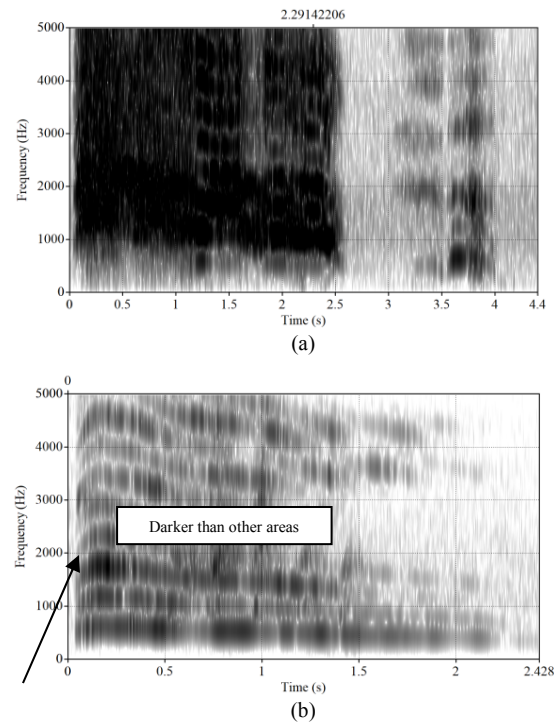


Fig. 4. Location of formants in (a) distress scream (b) joyful scream.

From Fig.4; it can be seen that the formants in the distress scream lie in the upper frequencies as compared to the joyful scream vocalization for a window length 0.025 and maximum formant up to 5500Hz. Fig. 5 shows the drastic difference between formant locations present for the first formant in both the vocal signals.

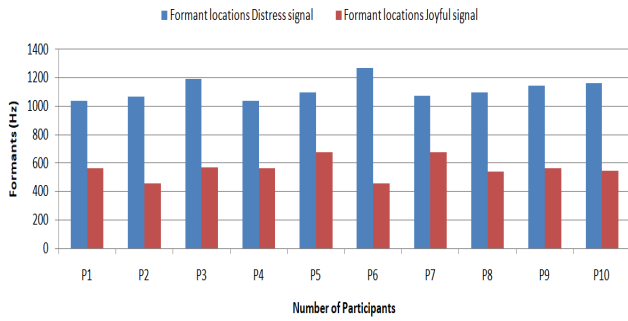


Fig. 5. Location of formants in distress scream and joyful vocalization

D. Signal Duration

To evaluate both high pitch signals, the signal duration is also an important analysis tool. Here in this study, we have considered the signal duration as the activity of the vocal tract emitted during the course of a single respiratory expiration and we have found that the distress scream signal has more duration than joyful scream. Table III shows the signal duration data for consideration.

TABLE III. COMPARISON OF SIGNAL DURATION FOR BOTH THE VOCALIZATION CATEGORIES

Participant	Signal Duration(Sec)	
	Scream	Hoot
P1	4.4	2.5
P2	6.2	2.8
P3	4.8	3.0
P4	4.3	2.1
P5	5.5	1.7
P6	4.9	1.9
P7	3.8	2.1
P8	4.8	2.6
P9	3.9	2.4
P10	4.1	2.2

It is observed from the Table III that the average signal duration of scream signal is 4.6 seconds whereas the hoot signal is 2.3 seconds.

IV. CONCLUSION AND FUTURE WORK

Although Joyful scream is also a high pitch vocalization yet there are differences while comparing it with a distress scream. We did not found many significant changes while computing F0 contour but it has been found that formant locations, sound intensity, and signal duration vary considerably. Moreover, there is more frequency content in distress scream as compared to joyful scream, which proves that even after falls into the category of high pitch vocalization, it has many different

characteristics as compared to distress scream. The future study will inculcate the in-depth analysis of other high pitch signals exists in-universe in the comparison of scream signals to find similarities and differences.

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